Project Design Phase

Proposed Solution Template

Date	27 September 2026
Team ID	
Project Name	Metro Ticket Generating System Using ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template:

The project team has filled in the following information for the proposed solution.

S.No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Traditional or partially manual metro ticket booking requires passengers or ticketing staff to repeat tasks such as selecting stations, working out the fare, recording ticket details and checking booking status. These steps can cause waiting time, inconsistent fare calculations and difficulty tracking booking records. A centralized, automated system is needed to manage bookings, calculate fares consistently, keep a unique identifier for every ticket and update booking status automatically.
2	**Idea / Solution description**	A scoped ServiceNow application that manages metro ticket bookings end to end: • Metro Station table: stores the stations (Central, City Square, Airport). • Metro Fare Rule table: stores the configured fare (base fare ₹10 + ₹5 additional fare per station). • Metro Tickets Booking table: stores ticket number, source, destination, passenger count, total fare, booking status and a QR Code Data field. • Business Rule: calculates the fare automatically from the selected journey. • Flow Designer flow “Metro Booking Confirmation”: when a booking is created, sets the booking status to Confirmed. In the reported test, a Central to Airport booking for one passenger returned a total fare of ₹20.
3	**Novelty / Uniqueness**	The project applies a low-code enterprise platform (ServiceNow) to a public-transport ticketing scenario, so no backend has to be built from scratch. Stations and fare values are kept as configurable records rather than fixed values, so they can be changed without rebuilding the application. Fare calculation and status updates run automatically through a Business Rule and an event-

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		driven flow. The QR Code Data field gives a ready place to add digital tickets later.
4	**Social Impact / Customer Satisfaction**	Automation reduces repetitive manual effort for ticketing staff and helps shorten the time needed to issue a booking. Passengers benefit from fares calculated by a consistent rule and from a unique ticket number that makes each booking easy to identify and track. As a prototype, it demonstrates how digital ticketing can reduce queues and errors; real-world impact would need a passenger-facing interface and operational testing.
5	**Business Model (Revenue Model)**	The current project is an academic prototype and does not generate revenue. Possible models if developed further: a subscription or licence for transit operators, campuses or shuttle services that need a booking system; a small per-transaction fee once payment integration is added; and paid customisation, integration and support services.
6	**Scalability of the Solution**	New stations and fare rules can be added as records without changing the application logic. ServiceNow provides the platform infrastructure for data storage, workflows and notifications. The same design can be adapted to bus, shuttle, campus and employee-transport bookings, and extended with QR-code tickets, notifications, payments and reporting dashboards. Production use would still require security, performance and load testing.